Semantifying Genomic Variant Data: VCF to RDF Conversion Framework

Elias Crum[1,2], Ruben Taelman[1], Bart Buelens[2], Gokhan Ertaylan[2], Ruben Verborgh[1]

[1]IDLab, Department of Electronics and Information Systems, Ghent University – imec, Belgium
[2]Flemish institute for Technological Research (VITO) Mol, Belgium

Abstract

Variant Call Format (VCF) files, the standard for representing patient genomic variant data, face limitations in interoperability, data linking, querying, and semantic interpretability. We propose a framework for converting VCF data into semantic data using the Resource Description Framework (RDF) to address these limitations. Our approach includes a comprehensive ontology, a storage-efficient RDF representation using Header Dictionary Triples (HDT), and integration with clinical metadata schemas like that proposed by SPHN. Our framework and the representation of VCF data semantically will contribute to greater integration, scalability, and usability of these genomic variant data in both genomic medicine and research.

Genome Data for Clinical Use

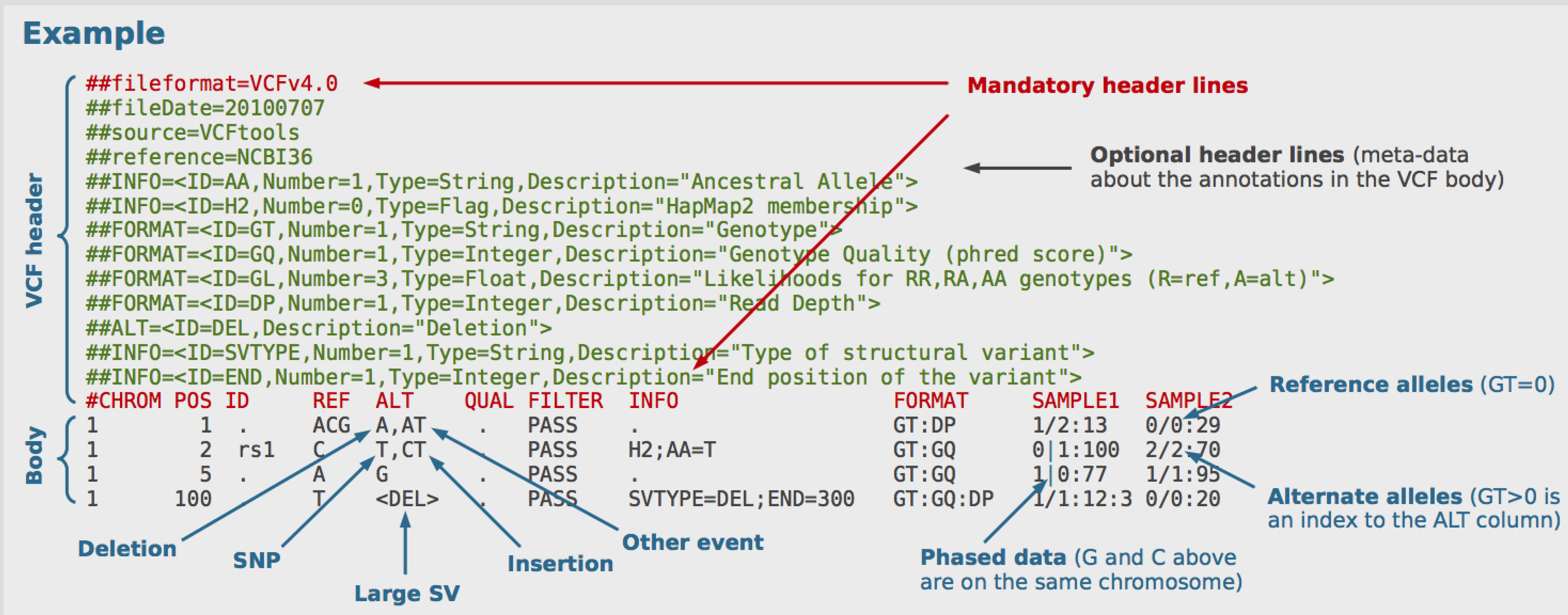
Predominantly VCF format

Objective

Represent VCF data semantically using RDF

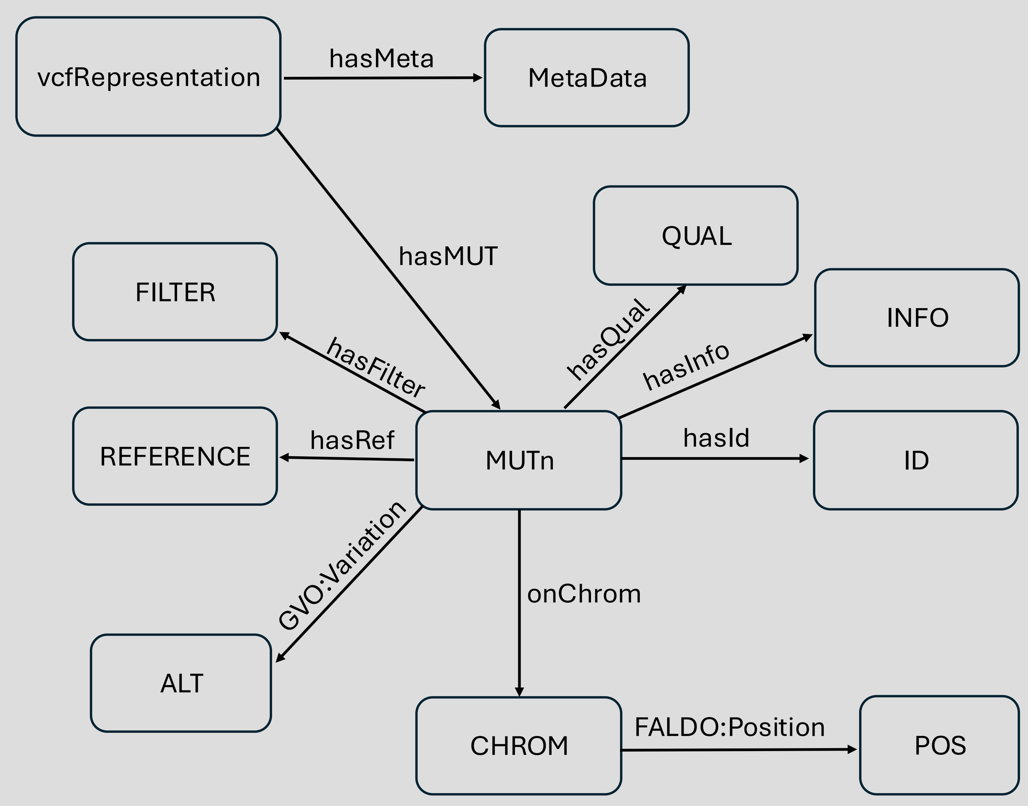
Proposed Framework

VCF Anatomy



https://davetang.github.io/learning_vcf_file/

VCF Ontology Visualization



FALDO – https://bioportal.bioontology.org/ontologies/FALDO
GVO -- http://genome-variation.org/resource/gvo

Conversion Framework

Existing Solutions:

- vcf2rdf -- https://github.com/togovar/vcf2rdf

PROBLEM: (extremely lightweight and does not represent all data within VCF file)

- SPARQLing-genomics -- https://gitlab.com/roelj/sparqling-genomics

PROBLEM: (uses proprietary vcf ontology + no longer maintained)

Proposed Solution:

- RDF Mapping Language (RML) -- https://github.com/RMLio/rmlmapper-java

WHY:
- NOT hardcoded into a conversion script
- Flexible (allows ontology evolution/extension)
- Established tool
- Offers methods for documentation generation

Future Directions

- Public Ontology Publication
- Pharmacogenomic health data linking
- Implementation Development

Contact

Funded by the Research Foundation - Flanders (FWO) [SB Fellowship 1S27825N]